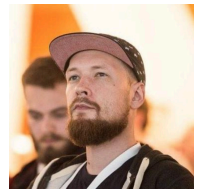


Sergey Rubanov

Software Engineer - AI Agent Runtimes, Web Runtimes & Developer Tools

Barcelona, Spain / hello@sergey.works / linkedin.com/in/chicoxyzyzzy



Professional summary

Software engineer, independent researcher, and open-source contributor focused on AI agent infrastructure, the web platform, and developer tools.

Contributes to TC39 and W3C standards work at the boundary between specification text, implementation behavior, and developer-facing APIs.

Core skills

AI / LLM systems

Agent orchestration, prompt/tool flows, model routing, eval & quality-regression checks, harness, llama.cpp, MCP/ACP

Runtimes / standards

ECMA-262, TC39, Test262, WebAssembly, JavaScriptCore, WebKit, browser/webview runtimes

Languages

TypeScript, Go, C++, Zig

Platform / observability

OpenTelemetry, traces/metrics/logs, Prometheus, Grafana, OTel Collector, Docker, Kubernetes, Terraform, AWS, GCP, Cloudflare

Selected Projects

Hecate — open-source AI gateway and agent-task runtime

<https://github.com/hecatehq/hecate>

Built an open-source local-first AI gateway and coding-agent console for routing OpenAI/Anthropic-compatible traffic across cloud and local models. Supports supervised external-agent sessions, task-backed agent runs, approvals, artifacts, MCP/ACP flows, provider diagnostics, usage/cost tracking, and OpenTelemetry.

Tech: Go, TypeScript, React, Tauri, OpenTelemetry, SQLite, MCP/ACP, OpenAI/Anthropic-compatible APIs.

Cynic — strict-only ECMAScript engine in Zig

<https://github.com/chicoxyzyzzy/cynic>

Built a strict-only ECMAScript engine in Zig for non-browser hosts such as edge runtimes, Workers, and server-side JS. Implemented parser/runtime pieces, bytecode-interpreter work, JIT - not yet (TBD), Test262-oriented conformance workflow, and runtime internals.

Tech: Zig, ECMAScript, Test262, compiler/runtime architecture.

Links

Personal website: sergey.works

GitHub: github.com

LinkedIn: linkedin.com

BlueSky: sergey.works

Twitter/X: x.com

Employment history

SEP 2025 – PRESENT

BARCELONA

Independent Software Engineer / Consultant

AI runtimes, web platform standards, developer tools, observability, and open-source runtime projects.

- Build AI/runtime and developer-tooling systems across LLM gateways, agent-task runtimes, web standards, JavaScript runtime internals, and observability.
- Design operator-facing reliability controls for agent systems: provider health/readiness, route diagnostics, trace links, usage/cost visibility, policy boundaries, approvals, and sandbox-aware execution.
- Contribute to TC39/ECMA-262 and Test262 by analyzing proposal trade-offs, refining specification text and semantics, and collaborating with contributors from browser vendors, JavaScript engines, and the broader standards community.
- Work independently in ambiguous, research-heavy spaces: turn loosely defined agent/runtime ideas into usable tools, documentation, release paths, and test loops.

MAY 2019 – PRESENT

Invited Expert, TC39

- Co-authored the Promise.any proposal and contributed editorial improvements to ECMA-262.
- Participate in proposal discussions, evaluate compatibility and ergonomics trade-offs, and develop Test262 tests to validate JavaScript behavior across engines.

MAY 2024 – MAY 2025

US, REMOTE

Software Engineer, Firetiger

Joined Firetiger after the acquisition of Dispatch.run; initially spent four months at Dispatch.run.

Dispatch.run:

- owned the CLI for the Durable Functions platform, improving developer experience and local-to-cloud workflow reliability.
- researched deployment approaches for running Go code through WebAssembly on Cloudflare.

Firetiger:

- contributed to the telemetry product across core features, infrastructure, and OpenTelemetry compatibility.
- supported early customers through onboarding, troubleshooting, integrations in Python/TypeScript, Vercel support, and Kubernetes/Terraform deployment workflows.

Stack: Go, Python, TypeScript, Node, Next.js, Vercel, Docker, Kubernetes, Terraform, AWS / GCP / Cloudflare, OpenTelemetry / Prometheus / Grafana / OTel Collector

MAR 2022 – JAN 2024

US, REMOTE

Software Engineer, Socket Supply Co

- Developed core Socket Runtime components in C++ to improve execution stability and platform behavior.
- Enhanced JavaScript APIs for Node.js compatibility and native capabilities across mobile and desktop runtimes.
- Built tooling to streamline developer workflows and reduce setup friction.

Stack: C++ / C / Objective-C++, JavaScript / TypeScript, Node.js, WebKit- and Chromium-based webviews: WKWebView, libwebkit2gtk, WebView2

DEC 2019 – SEP 2021

GENEVA, REMOTE

Software Engineer, RealAdvisor

- Developed a multi-tenant CRM and a custom document editor with layout templates to reduce manual formatting effort.
- Optimized the document rendering pipeline to improve load times and editor responsiveness.

Stack: React / Svelte, Flow, Node.js, Relay, Docker + Composer, Headless Chromium, GCP

DEC 2014 – AUG 2019

MALTA, REMOTE

Frontend Developer Lead, EXANTE

- Led development of a web trading terminal, Chrome extension, Ledger hardware-wallet applications, and web3/cryptocurrency projects.
- Mentored and interviewed web developers while leading delivery across React/Preact, TypeScript, RxJS, D3, Redux/MobX, and browser-extension work.

Stack: React/Preact, Redux/MobX, RxJS, D3, TypeScript, C

OCT 2013 – NOV 2014

MOSCOW, RUSSIA

Senior Developer, NPTV

- Built a React-Native-like JavaScript SDK for an interactive TV platform; helped adopt SpiderMonkey and later migrate to V8.
- Managed build systems, DevTools extensions, and R&D for client applications.

Stack: JavaScript, Babel, React, Chrome DevTools extensions

SEP 2012 – JUL 2013

MOSCOW, RUSSIA

Frontend Team Lead, KM Media

- Led frontend team to deliver frontend for an advertising platform.
- Established unit testing with Karma and Jasmine to reduce frontend defects and regression frequency.
- Coordinated with backend and product teams to resolve integration issues and improve data accuracy.

Stack: Angular, JavaScript

SEP 2009 – JUL 2013

ORYOL, RUSSIA

Senior Developer, Comfortel

- Built a CRM-like system with integrations for call center software (Genesys Agent Desktop, Oktell).
- Promoted to Team Lead in 2012. Led a team to implement a CRM system, enhancing client interactions and driving measurable improvements.
- Developed a dynamic web-based operator script generator, streamlining call center operations and reducing response times significantly.
- Managed and optimized databases (MySQL, MSSQL, Oracle), ensuring data integrity and enhancing system performance for critical business applications.

Stack: PHP, JavaScript, MSSQL, MySQL, OracleDB

Education

2009

ORYOL, RUSSIA

Master of Computer Science, Oryol State Technical University

Languages

Russian (Native), **English** (Proficient), **Spanish** (Novice)